THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The reaction $C_2H_4 + H_2 \longrightarrow C_2H_6$ with nickel is: 1

(a) addition reaction **(b)** combustion
(c) substitution reaction **(d)** esterification

2. Diamond does not conduct electricity because: 1

(a) no free electrons **(b)** it is soft
(c) it is metal **(d)** it is liquid

3. The molecular mass of the compound next to C_4H_{10} in homologous series is: 1

(a) 72 u **(b)** 86 u **(c)** 58 u **(d)** 44 u

4. $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ is: 1

(a) oxidation **(b)** substitution reaction
(c) combustion **(d)** addition reaction

5. Micelles are formed when soap is dissolved in: 1

(a) water **(b)** kerosene **(c)** oil **(d)** alcohol

6. The molecular formula of an alkane with 8 carbon atoms is: 1

(a) C_8H_{16} **(b)** C_8H_{14} **(c)** C_8H_{20} **(d)** C_8H_{18}

7. The compound formed on esterification of ethanoic acid with methanol is: 1

(a) methyl ethanoate **(b)** ethyl methanoate
(c) ethyl ethanoate **(d)** methyl methanoate

8. In a micelle, the hydrophobic tails point: 1

(a) upwards **(b)** outward in water
(c) inward towards oil **(d)** randomly

9. The molecular formula of an alkene with 5 carbon atoms is: 1

(a) C_5H_{10} **(b)** C_5H_8 **(c)** C_5H_6 **(d)** C_5H_{12}

10. Saponification of ethyl ethanoate with NaOH gives: 1

(a) sodium ethanoate and ethanol **(b)** sodium ethoxide and water
(c) ethene **(d)** ethanoic acid and ethanol

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by the term homologous series? State two features. 2

12. Why does soap not work well in hard water? 2

13. Write the electron dot structure of ethane and calculate number of shared pairs. 2

14. Why does ethanol burn with a clean flame while naphthalene burns with a sooty one? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw the structure of the three isomers of pentane and write names. 3

16. Write the equation for the saponification of ethyl ethanoate and give two products. 3

17. Why is ethanol toxic if consumed and what is denatured alcohol? 3

OR

An organic compound A of formula C_2H_6O reacts with sodium to give hydrogen. Identify A and write the equation. 3

18. Explain how addition of hydrogen to vegetable oil benefits and what is the catalyst. 3

OR

Write an equation for each: 3
(a) combustion of propane
(b) chlorination of ethane (one H).

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the reactions: ethanol -> ethene, ethanol -> ethanoic acid, ethanoic acid -> ester, with conditions. 5

OR

What is nomenclature? Name $CH_3CH_2CH_2OH$, CH_3CH_2CHO, CH_3CH_2COOH, $CH_3CH_2CH_2Cl$ and $CH_3COCH_2CH_3$. 5

20. How do soaps and detergents differ? Explain cleansing action and the effect of hard water. Give one drawback of detergents. 5

OR

Explain why ethanol and ethanoic acid differ in reaction with sodium, sodium carbonate, litmus, and pH. 5

END OF QUESTION PAPER — Total: 40 Marks

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